MULTIMODAL OPTIMIZATION OF PIEZOELECTRIC PATCH PLACEMENT AND SHUNT CIRCUIT DESIGN FOR ENHANCED STRUCTURAL VIBRATION CONTROL

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Thesis to obtain the Master of Science Degree in Mechanical Engineering

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Objectives

- The main objective is to build a passive control system to mitigate mechanical vibrations in composite materials.
- Studying the influence of patch pair division on vibration reduction attenuation.
- Analyzing system's response under low and high frequency conditions.
- Compare different control techniques and propose optimal control parameters.

Models

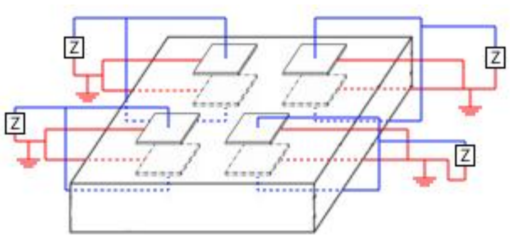

Figure 1: Example of multiple passive shunt control with 4 pairs of patches applied and an impedance Z with resistors and inductors.

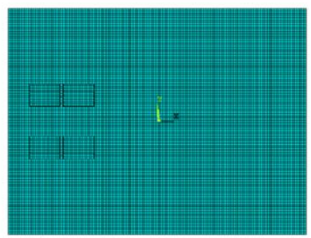

Figure 2: Example of an ANSYS mesh with 4 pairs of patches applied to the plate.

Results – Unimodal Approach

- Splitting up to 3 patches is efficient
- Single shunt isn't good for high frequency modes
- 1 patch pairs configuration is most efficient for low frequency modes

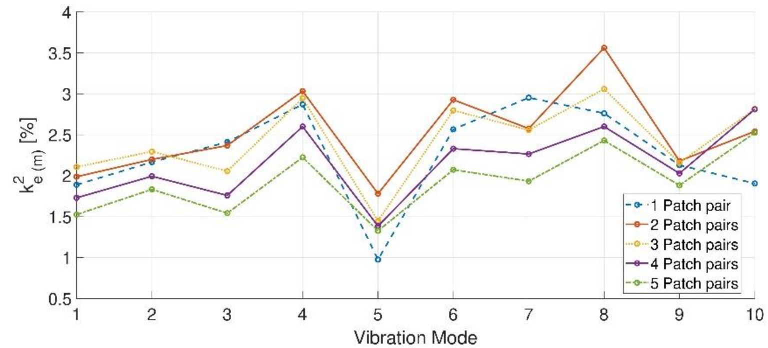

Figure 3: Optimal positioning of each number of patch pairs considering the unimodal approach.

Results – Multimodal Approach

- Multishunt circuit is the most efficient for multimodal applications
- 2 patch pairs was the best result for modes 6, 7 and 8

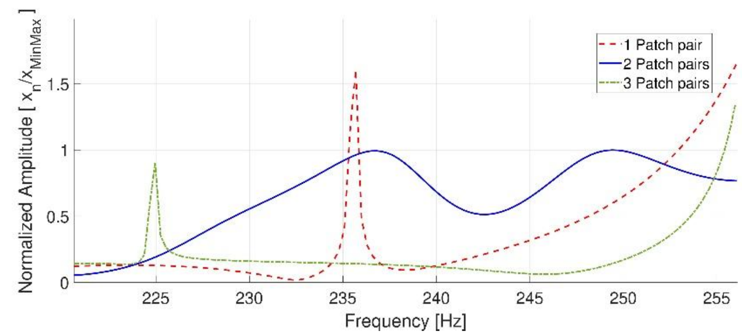

Figure 4: Analysis of the final result considering up to 3 pairs of patches in their ideal control parameters.

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